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Optimizing Sand Control for CCS Injection Wells: Innovations in Completion Strategies

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Abstract

Decarbonization of the energy sector urges a radical transformation to meet the net zero carbon emission target. In the last decades, many countries around the world have initiated or planned for the Carbon Capture and Storage (CCS) as one of the key projects aiming to permanently and environmentally safe storing of CO₂ underground. However, only a few projects have progressed into development and eventually into execution. Some of the biggest challenges are due to technical and commercial challenges to drive the success of the project. The objective of this paper is to focus on the CCS well construction, design consideration, and optimization diving deeply into sand control completion design. Two cases of offshore CCS projects in Asia Pacific are discussed in detailed design and planning phase.

CCS well construction part is similar to normal oil and gas extraction, but the main differences are designing injector well and injecting fluid is CO₂. Thus, understanding of CO₂ behavior, formation characteristic, and injection philosophy plays major roles for the completion design process to align with flow assurance study result, and how to cover the uncertainty with mitigation in-place.

Hence, these two case studies from Asia Pacific region illustrate the design considerations, challenges, and outcomes of implementing sand control solutions in CCS wells. The aim is to contribute to the evolving knowledge base and provide practical insights for future CCS well developments. The approach is on ensuring well integrity, optimizing injection performance, and cost optimization under challenging conditions. The study identified key challenges such as weak formations, permeability variations, and water hammer effects. Mitigation strategies included pressure equalization, stepwise shut-in approaches, and particle settling techniques. The completion designed to minimized sand production, maintained well integrity, and ensured long-term injectivity. The findings highlight the importance of tailored sand control solutions for the projects and the effectiveness of cased and perforated completions with rat holes for future workover flexibility.

Introduction

Carbon Capture and Storage (CCS) has emerged as a critical component of the global energy transition strategy, representing a new venture for the oil and gas industry. While the core competencies of subsurface engineering remain applicable, CCS projects introduce unique challenges that distinguish them from conventional hydrocarbon production operations. Among these, well completion design particularly sand control requires renewed attention due to the distinct operational and reservoir conditions inherent to CCS applications.

Despite growing interest and investment in CCS, publicly available technical references on sand control strategies specific to CCS wells remain limited. This knowledge gap is especially pronounced in the Asia Pacific region, where the deployment of CCS projects is accelerating. Traditional sand control design principles, largely derived from hydrocarbon-producing wells, may not be directly transferrable to CCS scenarios due to differences in reservoir drive mechanisms, fluid properties, and injection-driven well operations.

Unlike hydrocarbon wells, where production is driven by depletion and fluid withdrawal, CCS wells are designed primarily for the injection of supercritical CO₂ into subsurface formations. This fundamental shift in operating philosophy affects factors such as near-wellbore stress conditions, pressure regimes, and the potential for formation damage or sand production. As a result, sand control solutions must be tailored to accommodate these injection-centric conditions while maintaining long-term well integrity and injectivity.

Two CCS projects were simultaneously initiated in two different locations in the region under the operator's scope. One of the projects is located in offshore Malaysia, planned to be developed together with the development of a giant sour gas field. The gas production stream with CO₂ travels via a gas pipeline to an onshore gas plant. There, CO₂ will be captured and transported back to offshore platform via pipeline in a dense/supercritical phase and injected into depleted storage reservoirs as illustrated in [Figure 1\(a\)](#).

While the other project is in offshore Thailand where the CO₂ will be extracted from the production stream in CPP (Central Processing Platform) and injected back to depleted reservoir or aquifer through the CCS injector at the adjacent existing wellhead platform as shown in [Figure 1 \(b\)](#). The difference from one to the other is the location of CO₂ Compression & Dehydration System, whereby one is located offshore and the other onshore.

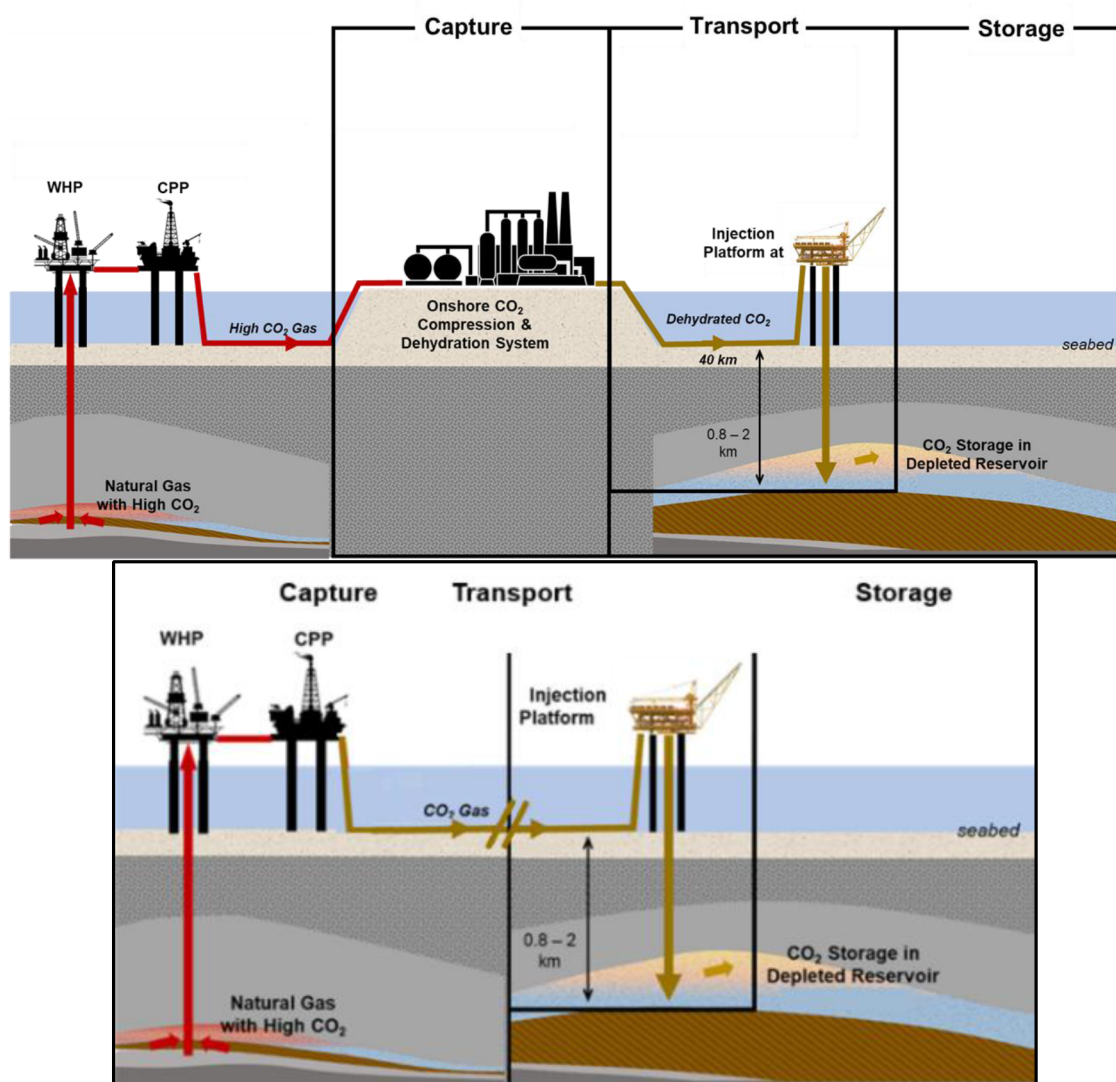


Figure 1—Offshore CCS Project Overview – CO₂ Extracted Onshore (Top) & CO₂ Extracted Offshore CPP (Bottom)

Literature Review

Upon literature review and reviewing sand control completion design in other CCS injectors globally including water injectors. The mechanism of how sanding occurs in a water injector is well covered by a paper written by Santarelli et al. The publication highlights an injector well that got shut in due to sanding and the sequence of four mechanisms that has been identified to explain the counterintuitive idea of sanding in injector well.

Weak or Unconsolidated Formation

Injecting condition in wells is basically putting force back into the formation, unlike in production drawdown. However, shut in condition could induce flowback and weak formations are prone to be produced as solid particles. Production historical records are a valuable information to predict and indicate if sand could still be produced in the injectors. Further sand prediction analysis (Figure 2) should be conducted to determine if the reservoirs are sand prone, especially during shut in condition. This is the first question for sand control completion to answer whether the formation will produce sand or not.

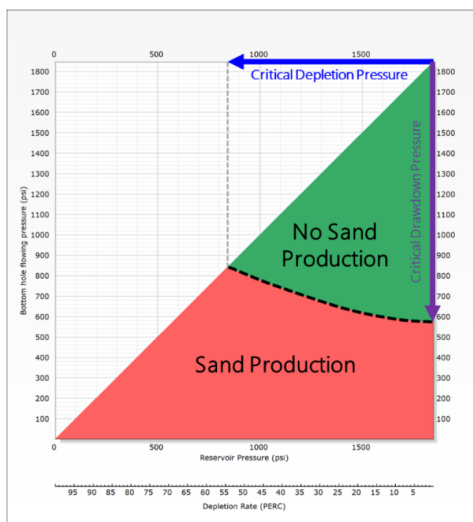


Figure 2—Example of sanding analysis

Cross Flow Due to Difference in Permeability and Pressure

Potential of crossflow that is caused by difference in permeability and pressure would carry the migrated fines. Azadbakht et al. exemplifies the impact crossflow can happen on sanding response of wellbores.

Depletion also plays a part in inducing forced crossflow. Example from Azadbakht et al. as shown in Figure 4 displays natural crossflow in S-2 that reach its peaks in two hours but becomes negligible in 36 hours. On the contrary, natural crossflow in depleted condition behaves like natural crossflow in the initial 6 hours but significant crossflow rate would persist to more than 24 hours due to difference in head.

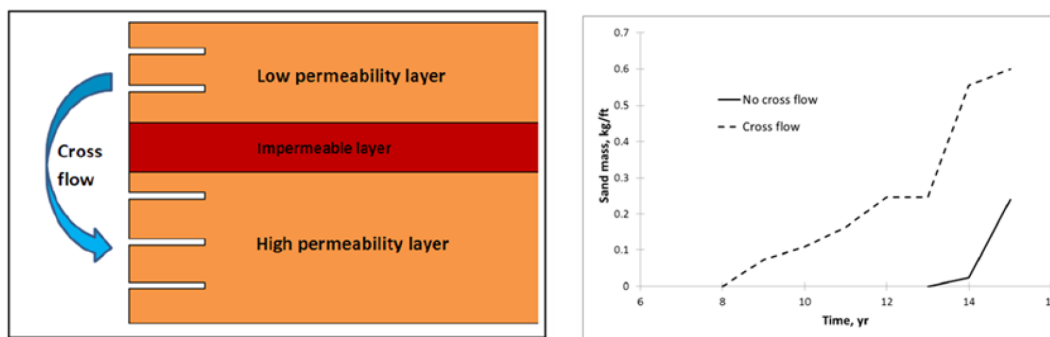


Figure 3—Effects of crossflow on sanding (Azadbakht et al.)

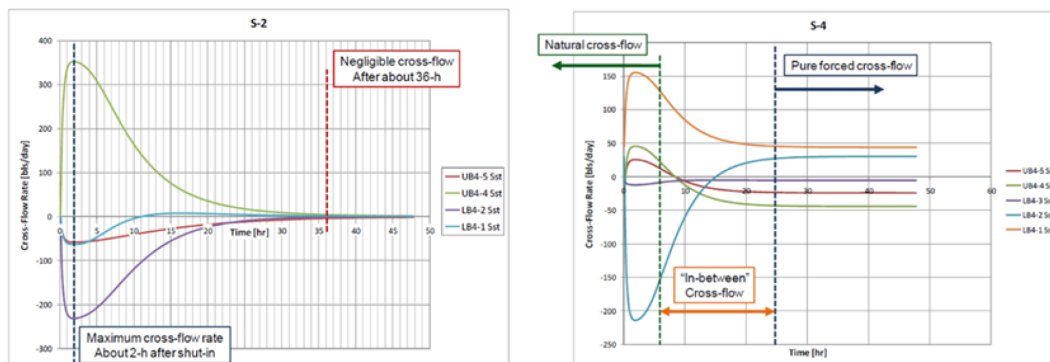


Figure 4—Crossflow in depleted sands (Azadbakht et al.)

Due to the potential for crossflow arising from variations in reservoir properties, reservoir engineers must optimize injection strategies to maximize CO₂ storage while minimizing the number of injectors. This often involves injecting into groups of reservoirs with similar injectivity and pressure characteristics in a commingled manner. Tripoppoom et al. conducted history matching for an offshore depleted gas field and utilized both reservoir simulation, proxy models and genetic algorithm to optimize the injection strategy, incorporating a combination of dedicated and commingled injection targets.

Water Hammer Effect from Well Shutting-in

Choi et al. describes the mechanism of water hammer as below:

- Upstream of the valve, when the valve is rapidly closed, the mass of water (that is moving forward) builds up to a high pressure as show in [figure 5](#) and shock wave at the position of the valve and pressure wave travels back and forth until energy dissipates due to friction.

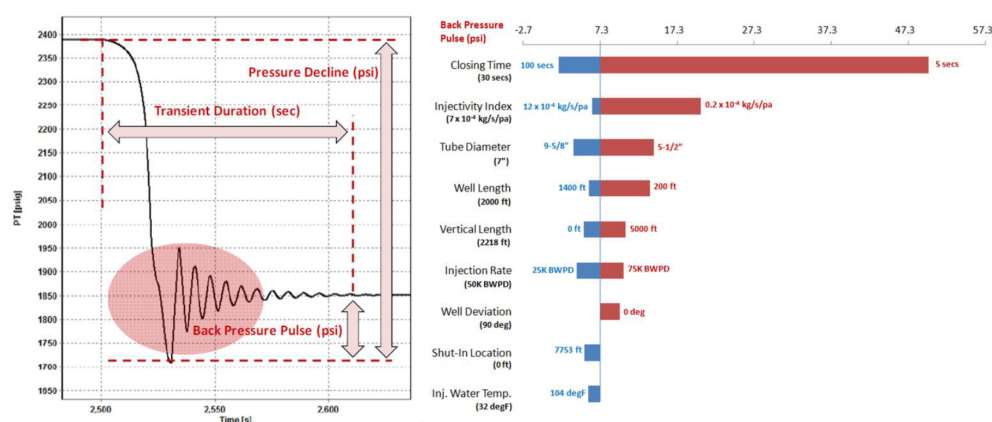


Figure 5—Water hammer effect (Choi et al.)

- Downstream of the valve, the mass of water tends to continue flowing by law of inertia, when the valve is instantly closed.

The instantaneous momentum is able to push the sand face to the point of failure. As shown in [figure 6](#), Wang et al. demonstrates the procedure of closing the valve results in how quickly or slowly the transient injection rate is changed from the initial injection rate to the complete shut-in state.

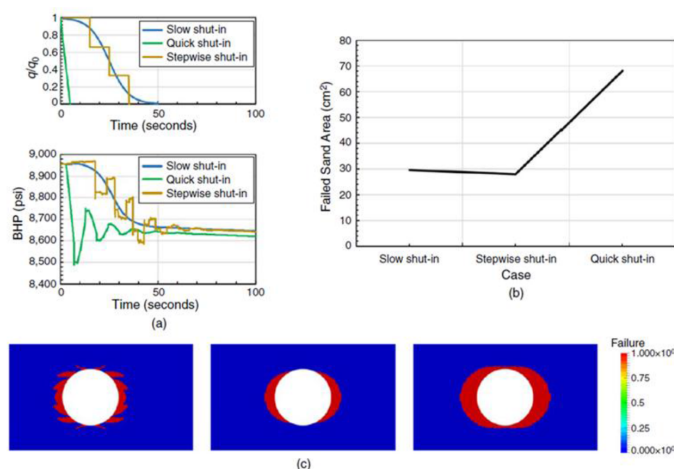


Figure 6—Effect of shut in transient condition on sanding (Wang et al.)

Floating Particulates

Fines might be flowed back into wellbore during shut-in and/or cross flow as explained above. This fine may plug formation pore throat once injection re-established as shown in Figure 7.

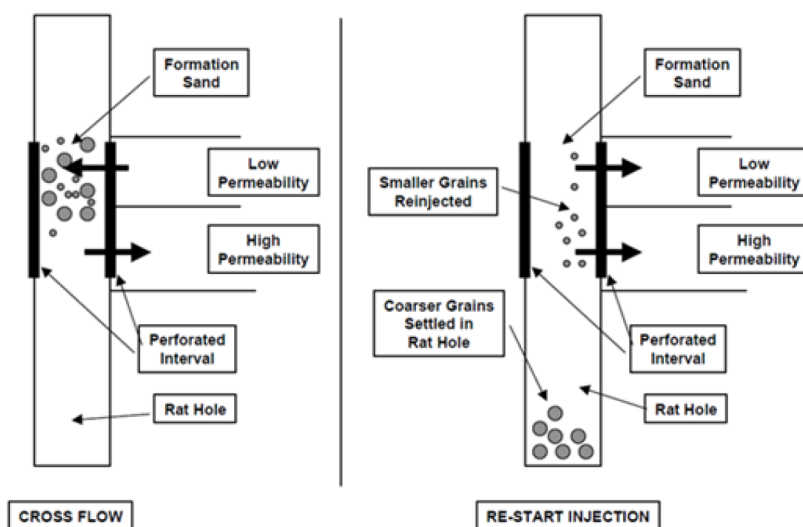


Figure 7—Schematic of perforation-tunnel plugging (Santarelli et al.)

A more comprehensive paper that discusses loss of injectivity is published by Sardpanah et al. In addition to the four mechanisms above, it also adds screen failure due to corrosion or erosion and thermal fractures.

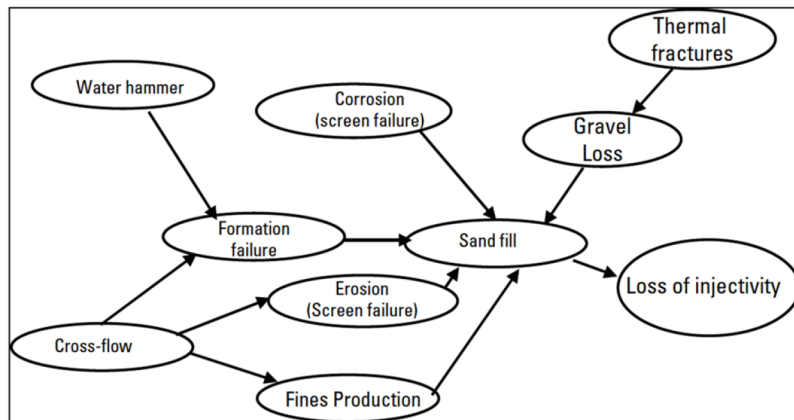


Figure 8—Mechanism of loss of injectivity (Sardpanah et al.)

Example of Sand Control Completion Design in CCS Injector

Below shows sand control completion in CO₂ injector, example from Peterhead CCS is completed with a 4" screen in its lower completion.

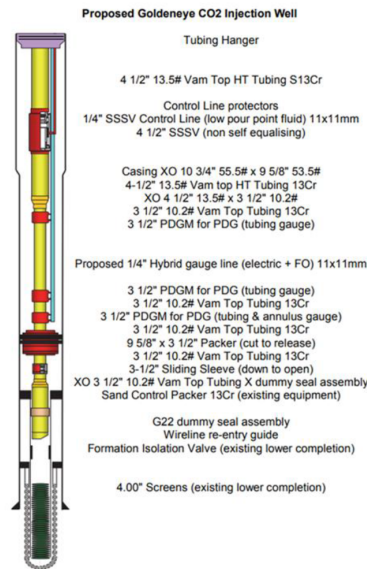


Figure 9—Example of Peterhead CCS Well Completed with Sand Screen, *Well Technical Specifications*

The other thought process with less conservative approach as seen in Acorn CCS, that adds some certain length of sump at the bottom of the well to accommodate sanding in the wellbore. This is the idea of sand management rather than sand control where the operator managed with well design and intervention if required.

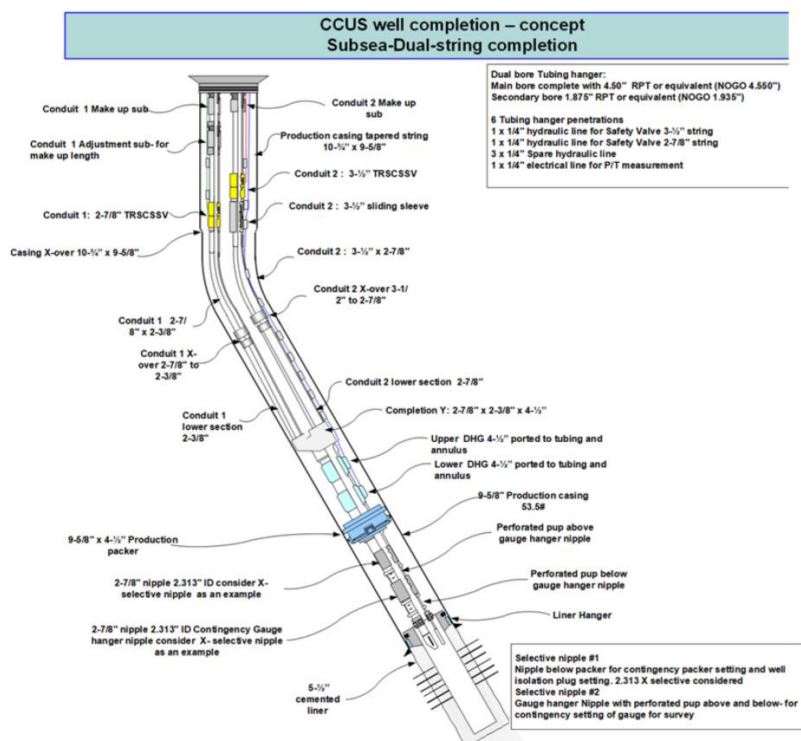


Figure 10—Example of Acorn CCS Injector, *Well Operating Guidelines*

The other sand control completion design seen in the Sleipner CCS project, in the Norwegian North Sea. There were initial problems with injectivity into the relatively unconsolidated Utsira sand formation due to sand influx which were solved by re-perforation the injection interval and installation of sand screen

and gravel packs (Hansen et al.). Leading to a well completion set-up in Figure 11 that has enabled steady injection of CO₂ for over 18 years. This evidence the risk of sanding persists even in CO₂ injectors.

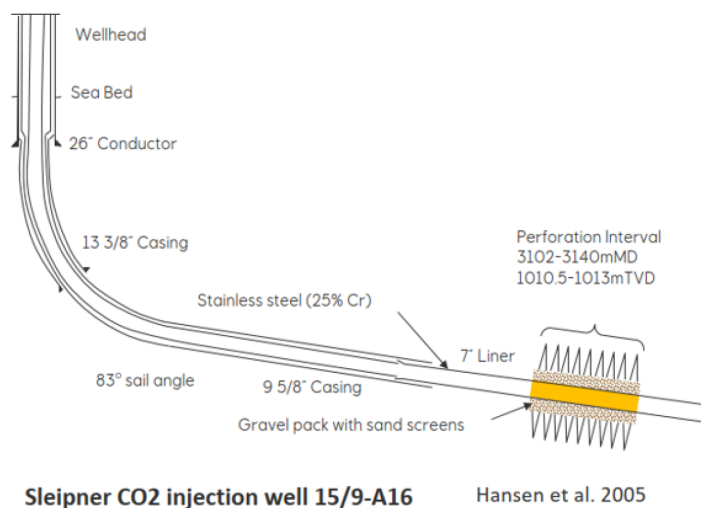


Figure 11—Example of Sleipner CCS Injectors

Sand Risk Study

Geomechanical Study

A geomechanical analysis was performed to evaluate the risk and severity of sand production during shut-in periods in CO₂ injection wells. The analysis utilized a three-dimensional Mechanical Earth Model (MEM), originally developed to support the broader CCS development study. This 3D-MEM provided direct access to key parameters influencing sand production potential, including as in-situ stress, rock strength, and pore pressure.

However, a key challenge emerged due to the absence of calibration data specific to sand production during shut-in events for the CO₂ injectors. To address this uncertainty, the study incorporated data from production wells located in the same reservoir units. These wells have documented sand production events, which were used to calibrate the model. By analyzing these analog wells, insights could be drawn and applied to improve predictions for the injection wells. One of the primary findings from the historical data indicated that sand production in producer wells typically began when the pressure drawdown exceeded 100 psi. Additionally, sand issues were more prevalent in reservoirs situated at depths shallower than 1000 meters (mTVDSS). These observations served as critical benchmarks in the calibration process and enhanced confidence in predicting sand behavior under shut-in conditions for the CO₂ injectors.

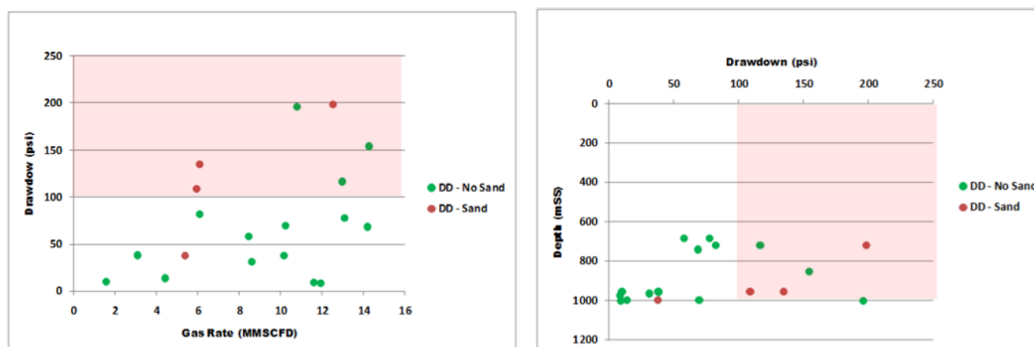


Figure 12—Sand production observed from producer wells

With the exception of grain size distribution, all other key parameters—such as elastic properties, rock strength characteristics, and in-situ stresses—were already well-calibrated using the 3D Mechanical Earth Model (MEM). The only remaining uncertainty in the geomechanical input data pertained to the grain size of the reservoir formations, which had not yet been directly calibrated. To address this, the pore pressure and in-situ stress conditions were extracted from a coupled 3D dynamic-geomechanical simulation model, specifically at the time step corresponding to when the producer wells were brought online. These extracted conditions served as inputs for a sand production analysis for producer wells. The sensitivity analysis was conducted on a range of plausible grain size distributions. The goal was to identify a set of parameters that would reproduce the observed sand production behavior in the field. Through iterative testing, a particular grain size distribution was found that successfully matched the documented sand production events in the producer wells.

Using this calibrated grain size distribution, the model was able to accurately replicate the observed sand production trends. Specifically, the model captured the onset of sand production in wells located in reservoirs shallower than 1000 meters (mTVDSS), which began producing sand when pressure drawdowns exceeded 100 psi. Conversely, the model also correctly predicted that wells completed in deeper reservoirs—those beyond 1000 mTVDSS—remained free of sand production under similar conditions, as illustrated in Figure 13.

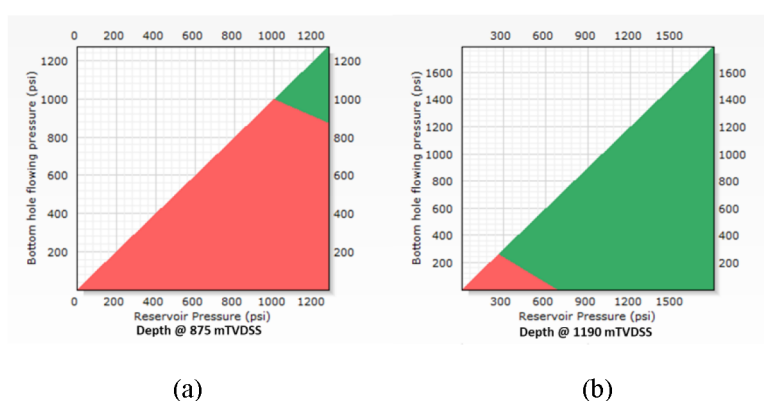


Figure 13—Sand production prediction from two producer wells (a) the well produce from reservoir shallower than 1000 mTVDss (b) the well produce from reservoir deeper than 1000 mTVDss

A similar modeling approach was extended to the CO₂ injection wells. However, for the injectors, the analysis utilized pore pressure and stress data extracted at two distinct phases of the injection process: the early stage and the late stage. These geomechanical conditions were taken from the coupled dynamic-geomechanical simulation model to reflect the evolving subsurface environment over time. As illustrated in Figure 14, the results for the injection wells were aligned closely with the trends observed in the producer wells. Specifically, reservoirs located at depths shallower than 1000 meters (mTVDSS) were found to be more prone to sand production, especially during the early stages of CO₂ injection. This heightened risk during early injection is primarily attributed to the significantly depleted pore pressure at the end of the field's production life. As a result, the effective stress acting on the reservoir rock is elevated, increasing the likelihood of sand mobilization and production. In contrast, during the later stages of injection, the gradual buildup of pore pressure—along with a partial rise in total stress—reduces the net effective stress. This change in stress conditions shifts the reservoir toward a more stable mechanical state, effectively lowering the risk of sand production as injection continues.

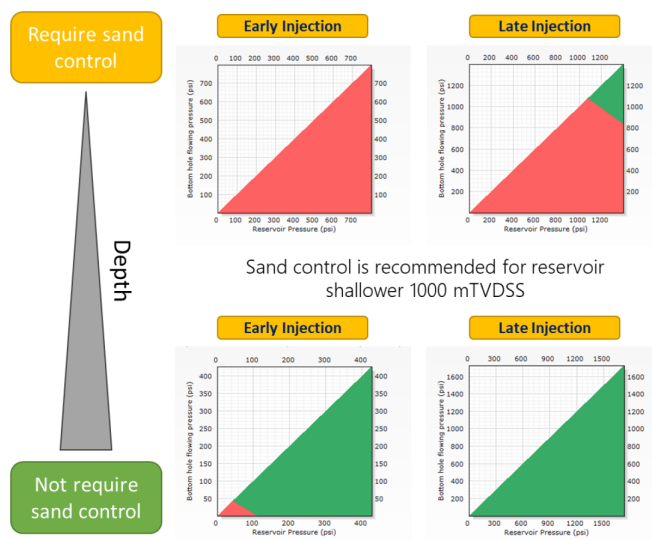


Figure 14—The sand production prediction in injector wells

Crossflow Analysis

Refer to the offshore Malaysian field in this study, crossflow behavior during CO₂ injection was investigated. A compositional reservoir model, calibrated with historical field data, was employed to predict crossflow among groups of small reservoirs planned for commingled injection. Two reservoir groups were analyzed: Group G1, comprising five reservoirs (Res 1.1 to Res 1.5), and Group G2 comprising two (Res 2.1 and Res 2.2).

Initially, two operation scenarios were compared: (1) immediate injection following perforation, and (2) perforation followed by a one-month shut-in period before injection. As shown in Figure 15 and Figure 16, initiating CO₂ injection after one-month shut-in period reduced crossflow in both G1 and G2 groups.

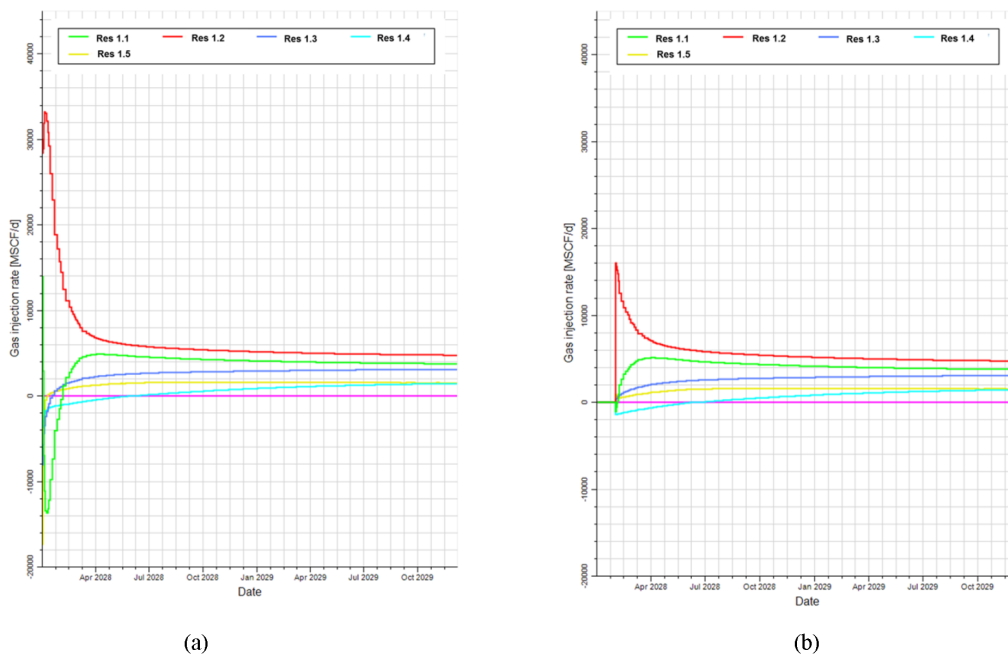


Figure 15—CO₂ injection rate by reservoir in Group G1: (a) Immediate injection following perforation and (b) one-month shut-in before injection

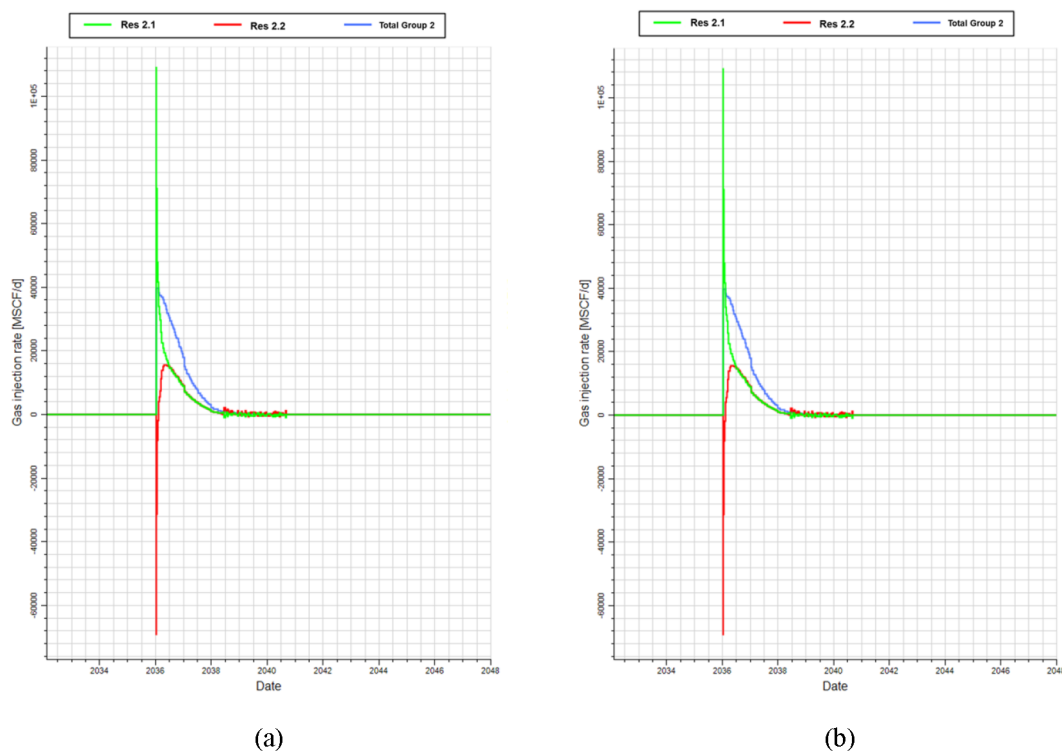
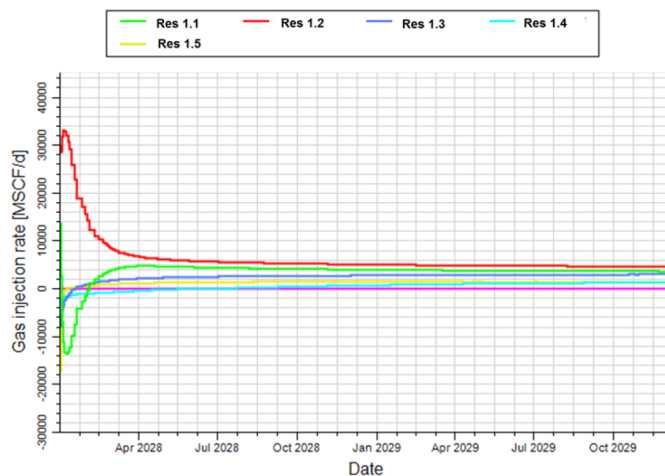
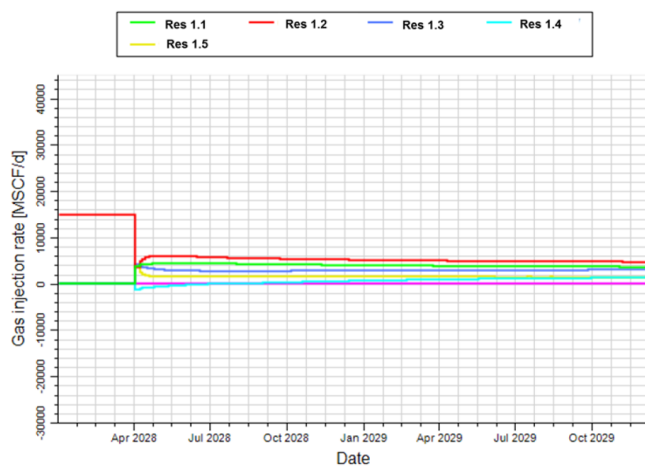


Figure 16—CO₂ injection rate by reservoir in Group G2: (a) Immediate injection following perforation and (b) one-month shut-in before injection

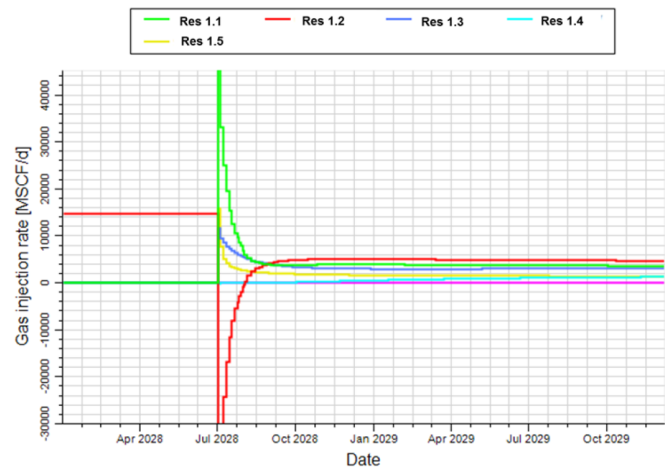
To further mitigate crossflow, staged perforation strategies were evaluated. For Group G1, two strategies were simulated: (a) a 3-month delay and (b) a 6-month delay between the initial perforation/injection in Res 1.2 and subsequent perforations of other reservoirs. Results indicate that the 3-month staged approach is optimal. In contrast, the 6-month delay led to reverse crossflow from Res 1.2 into the other reservoirs, as illustrated in Figure 17.



(a)



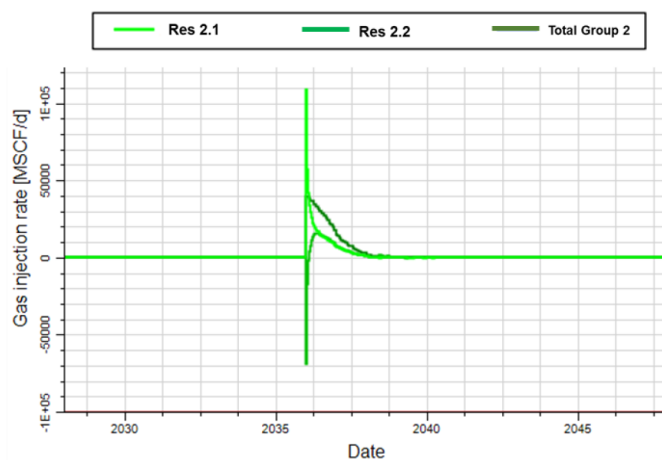
(b)



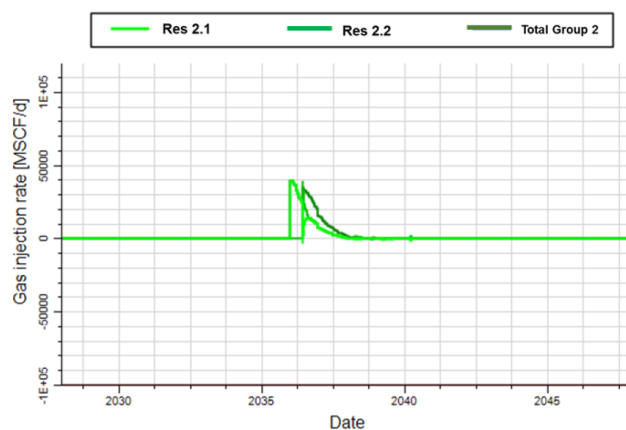
(c)

Figure 17—CO₂ injection rate by reservoir of stage perforation in Group G1: (a) No stage perforation, (b) 3-month staged perforation, and (c) 6-month staged perforation

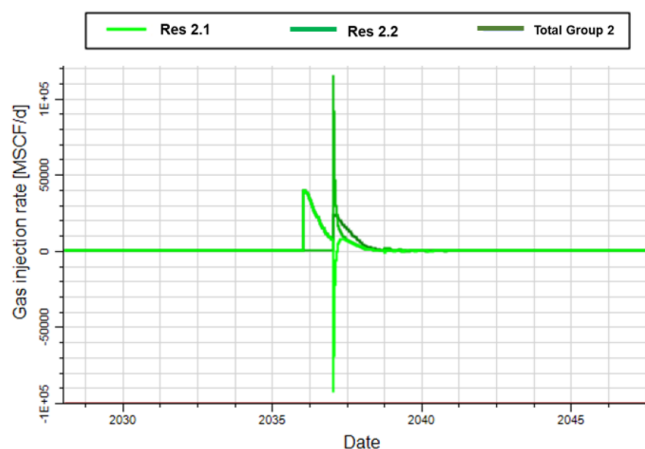
A similar analysis was conducted for Group G2, assessing 6-month and 12-month staged perforations. The 6-month delay yields more favorable results, whereas the 12-month delay induced reverse crossflow from Res 2.1 into Res 2.2 (Figure 18).



(a)



(b)



(c)

Figure 18—CO₂ injection rate by reservoir of stage perforation in Group G2: (a) No stage perforation, (b) 6-month staged perforation, and (c) 12-month staged perforation

In conclusion, the reservoir simulation successfully predicted crossflow behavior under various operational scenarios. These findings provide critical insights for field implementation, enabling strategies to minimize crossflow and, consequently, reduce the risk of sand production during actual CO₂ injection operations.

Water Hammer Effect

Transient simulation from one of the CO₂ injection reservoir has shown maximum pressure fluctuations of approximately 8 bars (116 psi) modelled for the early life of the well as shown in Figure 19. Although the operational pressure in early life is lower (50 bar), due to lower operating temperature in early life (12°C), the fluid is denser than late life with higher pressure and higher temperature (85 bar & 23°C). Therefore, water hammer would be slightly higher in early life compared to late life. The risk of sand production caused by water hammer effect can be assessed and controlled by combining the sand production prediction analysis.

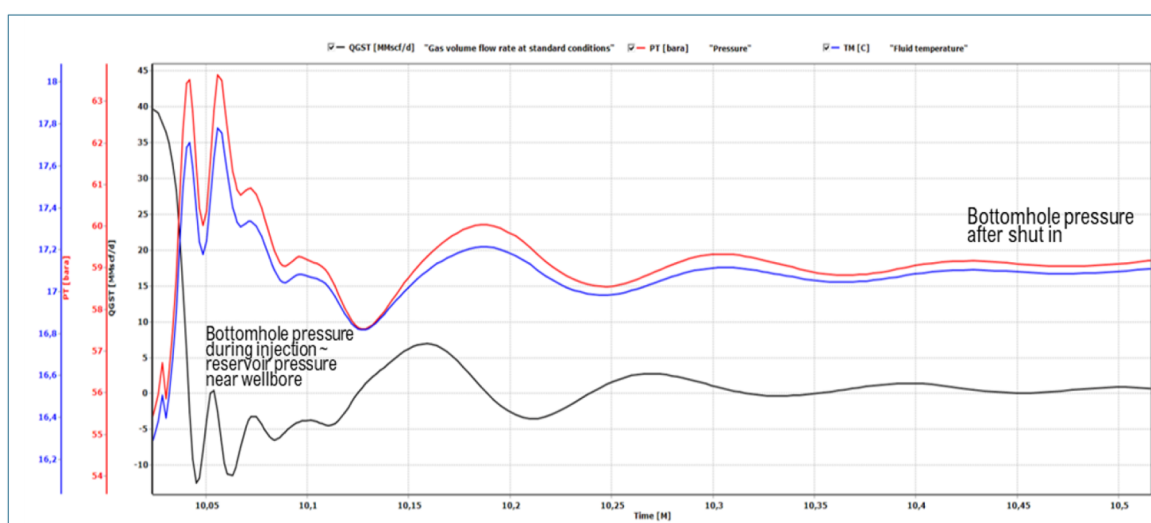


Figure 19—Water hammer effect simulation using dense CO₂ – Example in one of the injections well

Completion Design

Moving Beyond Conventional Gravel Pack in Depleted Formations

Particle Size Distribution (PSD) analysis from offset wells and core data initially suggested that a gravel pack completion would be a best suit sand control solution, (Tiffin's criteria for gravel sizing). However, a critical consideration for the target CCS injectors was the reservoir's depleted state.

The gravel pack placement simulations, utilizing this reservoir data, provided critical insights. These simulations Figure 20 explicitly demonstrated a high likelihood of achieving only a suboptimal pack across the intended completion interval. Specifically, within the context of the depleted reservoir pressures and the significantly lower fracture gradient of the second sand interval (Sand 2), the simulations showed a pronounced risk of premature screen out before a complete and effective gravel pack could be placed.

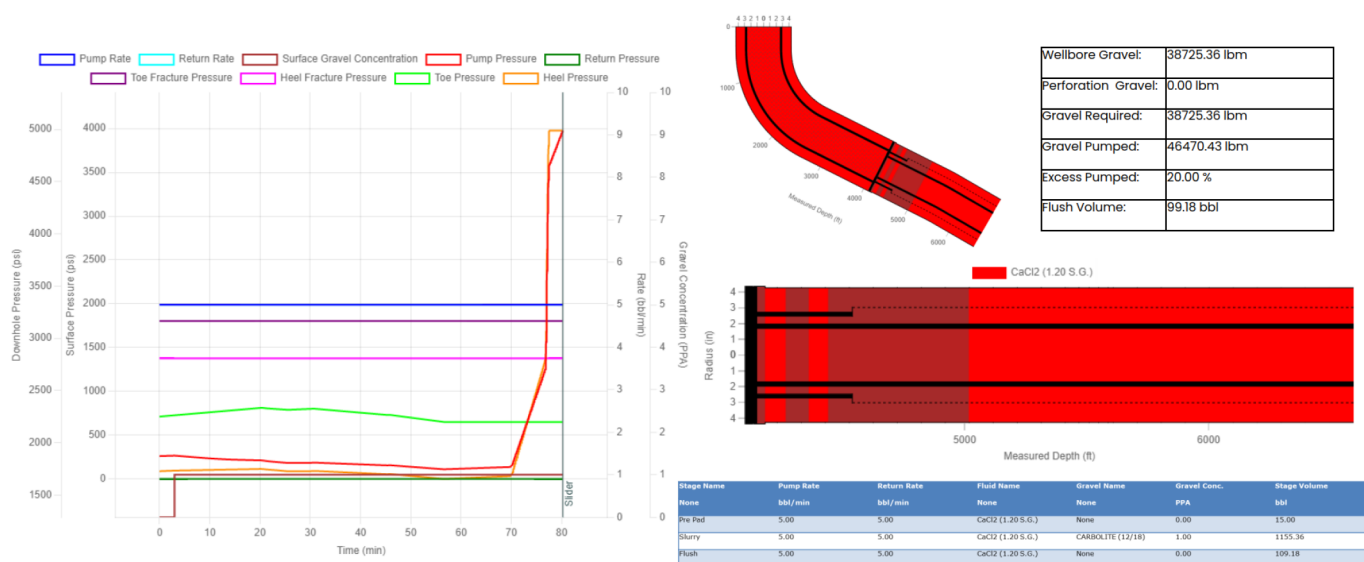


Figure 20—Open hole gravel pack simulation results

Adding to these challenges, the formation's non-uniform grain size distribution limited the applicability of most standard screen types. This heterogeneity further complicated the design of any direct sand exclusion method at the formation face.

Flexibility of Well Design and Solids Management

The initial well architecture featured a cased hole with a liner section, a design strategically chosen to facilitate the secure installation of essential Measurement, Monitoring, and Verification (MMV) equipment and to optimize costs by selectively using Corrosion Resistant Alloys (CRA) only for the liner exposed to the CO₂ injection zone. Recognizing that these specific design requirements for the liner, when combined with the inherent challenges of gravel packing in depleted reservoirs—particularly the risk of gravel displacement—rendered conventional sand control problematic, an alternative lower completion strategy was adopted. The operator therefore developed a cased and perforated design incorporating a significant sump, or "rat hole," below the perforated interval. This design philosophy shifts from complete near-wellbore sand exclusion to a strategy of managing produced solids that may enter the wellbore.

The key advantages influencing this decision were:

1. **Enhanced Operational Flexibility:** A cased and perforated completion offers significantly greater flexibility for future well interventions and workovers compared to a gravel-packed completion. Remedial operations, data acquisition, or re-perforation are considerably less complex and costly.
2. **Mitigation of Premature Screen out Risks:** This design entirely avoids the risks of premature screen out and incomplete packing associated with gravel pack operations in depleted formations.
3. **Accommodation of Produced Solids:** A dedicated rat hole provides a designated space for produced formation materials to accumulate over time, preventing them from immediately obstructing perforations or damaging downhole equipment.

Based on historical production performance and solids accumulation rates observed in analogous wells in the same field, a 50-meter rat hole in 7" liner section was incorporated into the well design. This length was determined from the amount of sanding records in producers. The rat hole offers sufficient volumetric capacity to accommodate the anticipated sand and fines production over an extended operational period up to period of 2 years of sanding occur into wellbore during shut-in or cross flow, thereby minimizing the frequency of required cleanout operations.

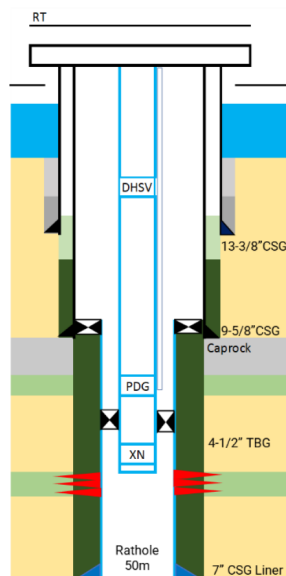


Figure 21—CCS injector completion schematic

Operational Philosophy: Managing Sand Mobilization and Settling During Shut-In

A key operational consideration for the cased and perforated completion with a rat hole is the management of formation fines and sand particles that may be mobilized during shut in and flowback periods. If these suspended particles are not adequately managed, they can repopulate the near-perforation area or the perforations themselves. Upon restarting injection after a shut-in period, these particles risk immediate mobilization, potentially leading to perforation plugging, increased skin damage, and impaired injectivity.

To mitigate this, the operational philosophy incorporates planned shut-in durations specifically designed to allow mobilized particles to settle gravitationally out of the critical near-wellbore region and into the 50-meter rat hole. The determination of the required settling time hinges on identifying the particle sizes most likely to cause plugging.

Analysis based on established particle bridging criteria to particle diameter ratio, D/d , in the range of 2.4 loose pack to 6.4 tight pack, indicates that particles in the size range of approximately 33 to 91 microns are most critical for causing plugging within the formation face in this field context ($D_{50} = 210$ micron). The required 6-hour settling period was established based on Stokes' Law, which is primarily accurate for particles smaller than 110 microns, to estimate the time necessary for settling of particles within this critical size range.

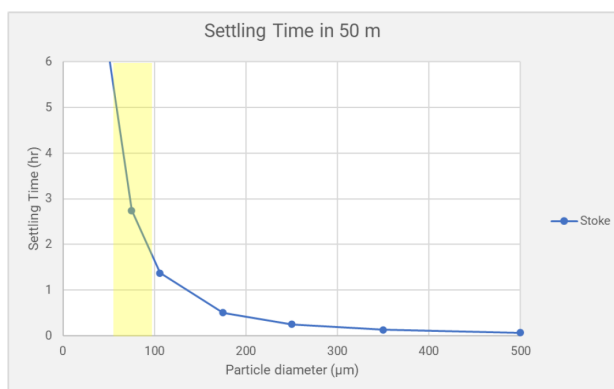


Figure 22—Settling time of sand in 50m rat hole by Stoke's equation

Flexibility of the Design

Cased & perforation in 7" liner section with 50m rat hole has also advantage on its high flexibility for future workover jobs compared to gravel pack completion from day 1. This design opens opportunity for cased hole gravel pack in 7" section in the future if sand problem arises beyond rat hole strategy. Additionally, the choice of a 50m rat-hole was made to accommodate particle accumulation, based on historical records of the production wells in particular field. Shut in and wait for sand settling time for 6 hrs. are captured in operation philosophy of shallow reservoirs.

The same thought process applied to the other CCS project where 4-1/2" completion string which will be installed after completed 7" liner will be stabbed in the sealbore assembly. This is due to the core from the injection sand dictates high fine. UCS is in the range of 1,500 psi which sand prone as per field history. Hence, the plan is to collect core sample in next drilling campaign and this design maintains flexibility to convert lower completion to be gravel pack completion in the future if required.

Conclusion

Designing sand control for CCS injector wells presents unique challenges that diverge from conventional hydrocarbon production wells. Key considerations include the risk of sand production during shut-in and crossflow conditions, formation heterogeneity, reservoir depletion, and the effect of water hammer during transient operations. Geomechanical modeling revealed higher sand production potential in shallow, depleted reservoirs with weak formations. Additionally, flow assurance, crossflow simulations, and particle settling behavior were factored into the completion design to ensure injectivity and integrity over the well's lifecycle.

The selected completion design is a cased and perforated wellbore configuration with a 50-meter rat hole in a 7" liner section in both projects. This configuration moves away from conventional gravel pack approaches, which were deemed suboptimal due to reservoir depletion and heterogenous grain size. The rat hole design supports solids management rather than full exclusion, allowing produced sand and fines to settle below the perforations during shut-in periods. Settling time analysis based on particle bridging and Stokes' Law established a 6-hour operational delay before resuming injection. This practical design aligns with the field's formation properties and enables controlled sand handling without compromising well performance.

Both projects carry flexibility, operational advantages, and cost saving from this completion strategy. The cased and perforated system facilitates future re-perforation, well intervention, and even conversion to gravel pack if required, minimizing long-term costs and downtime. The rat hole provides a buffer zone for sand accumulation, reducing the risk of near-wellbore plugging and maintaining injectivity. Moreover, the modular approach allows future adaptability across different CCS projects, especially in reservoirs with variable rock strength and sand-prone conditions. Overall, the design balances operational practicality with technical resilience, making it a robust and scalable solution for CCS well development.

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